

Fraud & Transaction Risk Intelligence

End-to-End Fraud Detection, Risk Analysis & Business Intelligence

PROJECT TYPE

Data Analytics | Machine Learning | Risk Intelligence

CORE TECHNOLOGIES

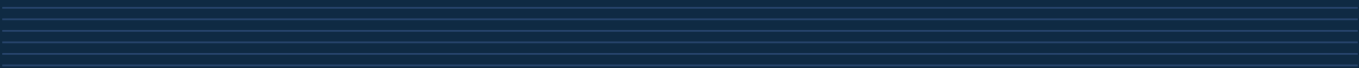
Python • SQL • PostgreSQL • Power BI • Scikit-learn

DELIVERABLES

ML Fraud Classifier • SQL Risk Analysis • Power BI Dashboard

REPOSITORY

GitHub Project — [github.com/\[your-username\]/fraud-transaction-risk-intelligence](#)



02 Executive Summary

This project builds an end-to-end fraud and transaction risk intelligence workflow on the IEEE-CIS Fraud Detection dataset, combining data engineering, exploratory analysis, machine learning, SQL-based risk analysis, and business intelligence reporting into a single coherent pipeline.

The core challenge is that fraudulent transactions are rare compared to legitimate ones, which makes fraud detection a **class-imbalance problem** rather than a standard classification task. In this dataset, fraud accounts for roughly 2.4% of all transactions, so a model that simply predicts “not fraud” for everything would already be over 97% accurate while catching zero fraud — which is why accuracy alone is not a meaningful evaluation metric here.

The project addresses this by preparing and joining transaction and identity data, engineering 21 behavioural and risk-related features, and training and comparing two classification models: Logistic Regression and Random Forest. Models were evaluated using metrics suited to imbalanced data — precision, recall, F1, ROC-AUC, and PR-AUC — rather than accuracy alone. **Random Forest was selected as the strongest overall model based on PR-AUC (0.1996)**, giving the best balance of precision and recall for identifying suspicious transactions.

Beyond modelling, PostgreSQL was used to independently query the transaction data for data-quality checks, behavioural patterns, and risk segmentation, while a Power BI dashboard consolidates transaction volume, fraud rate, card, and product-level breakdowns for interactive business exploration. Together, the three layers — ML, SQL, and BI — turn a raw transaction dataset into a workflow that mirrors how a real risk team would approach fraud prioritisation.



03 Business Problem

Financial institutions process enormous volumes of transactions, and only a very small fraction are fraudulent. This creates several practical challenges that shaped the approach taken in this project:

Fraud is a needle-in-a-haystack problem

With fraud representing roughly 2–3% of transactions, the vast majority of the data offers little direct signal. Models can easily achieve high accuracy by ignoring fraud entirely, which is why accuracy is a misleading metric for this use case.

Precision and recall trade off against each other

A model tuned purely for recall will flag many legitimate transactions as fraud (costly to investigate); a model tuned purely for precision will miss real fraud cases (costly to the business). Fraud detection therefore requires evaluating models on metrics such as F1, ROC-AUC, and especially PR-AUC, which better reflect performance on the rare positive class.

Investigation capacity is limited

Risk and fraud teams cannot manually review every transaction. A model or dashboard that ranks transactions by suspicion — rather than just labelling them — helps investigators prioritise the cases most worth their limited time.

Behavioural signals matter as much as raw values

Patterns such as missing identity or device information, unusual transaction amounts relative to a customer's typical behaviour, and card/product combinations can be as informative as the transaction amount itself. This motivated the feature-engineering work described in Section 7.

In short, this project treats fraud detection as a **risk-prioritisation problem**: the goal is not a perfect classifier, but a system that surfaces the transactions most worth a human's attention, supported by SQL-based behavioural analysis and a dashboard for ongoing monitoring.

04 Project Objectives

- Prepare and clean transaction and identity data from the IEEE-CIS dataset.
- Explore fraudulent transaction patterns through structured EDA.
- Engineer meaningful transaction and risk-related features.
- Build and compare fraud classification models (Logistic Regression, Random Forest).
- Evaluate models using metrics appropriate for imbalanced fraud data (precision, recall, F1, ROC-AUC, PR-AUC).
- Analyse transaction behaviour independently using SQL / PostgreSQL.
- Segment transactions according to risk-related patterns for investigation prioritisation.
- Communicate findings through an interactive Power BI dashboard.

05 Dataset & Data Preparation

The project uses the **IEEE-CIS Fraud Detection dataset**, a widely used industry benchmark for transaction-level fraud modelling. It is split across separate transaction and identity files, which were joined on a shared key prior to analysis and modelling.

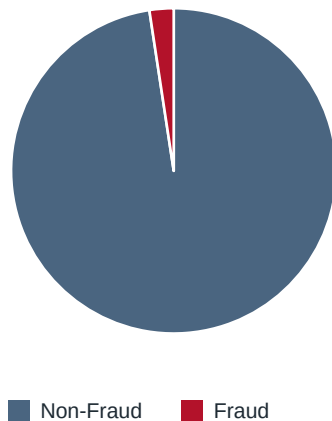
Component	Description
train_transaction.csv	Transaction-level training data (amount, product, card, address, timing, engineered fields, target label).
train_identity.csv	Device, browser, and identity-related attributes linked to a subset of transactions.
test_transaction.csv / test_identity.csv	Held-out transaction and identity data used for evaluating generalisation.
sample_submission.csv	Reference submission format matching the original competition structure.
Target	isFraud — binary label indicating a fraudulent transaction.
Join key	TransactionID — used to merge transaction and identity records.

Data Preparation Steps

- Loaded and merged the transaction and identity tables on **TransactionID**, keeping transactions without identity records (identity coverage is partial in this dataset, which itself carries signal — see Section 7).
- Assessed missing-value patterns across transaction and identity columns to inform feature engineering rather than blanket removal of incomplete rows.
- Standardised categorical fields used downstream (e.g. **ProductCD**, **card4**, **card6**, **DeviceType**, **P_emaildomain**, **R_emaildomain**) for consistent encoding in the modelling pipeline.
- Derived time-based fields from the raw transaction timestamp (hour, day, week) to support both EDA and modelling.
- Preserved the natural class imbalance in the data (no synthetic oversampling), and instead addressed it through model choice and evaluation metrics.

06 Exploratory Data Analysis & Fraud Patterns

Exploratory analysis focused on understanding where fraud concentrates, rather than exhaustively profiling every column. The dashboard in Section 12 reflects the same underlying distribution used during EDA: fraud accounts for a small minority of transactions overall.



Class imbalance. Fraudulent transactions make up a small share of total volume (2.4% fraud vs. 97.6% non-fraud in the analysed data). This confirms that any modelling approach must be evaluated on imbalance-aware metrics, and that a purely accuracy-driven model would be misleading in production.

Transaction amount behaviour. Fraudulent and legitimate transactions differ in typical amount and in how much an individual transaction deviates from a customer's usual spend — which is why amount-deviation features (e.g. AmountVsMedian, AmountZScore) were engineered rather than relying on raw amount alone.

Product category patterns. Fraud is not distributed evenly across product categories (ProductCD); certain categories carry disproportionately more fraud volume, which supports category-aware risk segmentation (Section 11).

Additional Patterns Investigated

- **Card-related patterns** — card type and network (card4/card6) show uneven fraud concentration, consistent with the card-level breakdown surfaced on the dashboard.
- **Device-related information** — device type and presence of device metadata differ between fraud and non-fraud transactions, motivating HasDeviceInfo as an engineered signal.
- **Missing identity information** — many transactions lack linked identity data; whether identity information exists at all (HasIdentityInfo) is itself a useful behavioural signal, not just a data-quality issue.
- **Time-related behaviour** — transaction hour, day, and week were examined for concentration effects, supporting the time-based features used in modelling.

Business meaning: rather than treating these as isolated statistical differences, each pattern was framed as a monitoring signal — e.g. missing device/identity metadata combined with an atypical transaction amount is a stronger risk indicator than either factor alone, which is the logic carried into feature engineering and later into SQL-based risk segmentation.

07 Feature Engineering

The final modelling feature set contains **21 features**, grouped below by the type of signal they capture. Grouping reflects the reasoning behind the feature set rather than a flat column list.

Group	Features	Why it's useful
Transaction Behaviour	TransactionAmt, TransactionHour, TransactionDay, TransactionWeek, LogTransactionAmt, TransactionAmtDecimals, AmountVsMedian, AmountZScore	Captures spend size, timing, and how unusual an amount is relative to typical behaviour — core signals for spotting atypical activity.
Missingness & Identity Signals	MissingFeatureCount, MissingFeatureRatio, IdentityFeatureCount, HasIdentityInfo, HasDeviceInfo	Treats data completeness itself as a behavioural signal; fraudulent sessions often carry different missingness patterns than legitimate ones.
Card & Address Signals	CardFeatureCount, AddressFeatureCount, card4, card6	Encodes card network/type and how much card/address metadata is available, both of which vary between fraud and legitimate transactions.
Categorical Transaction Information	ProductCD, DeviceType, P_emaildomain, R_emaildomain	Captures product context and device/email characteristics, which show uneven fraud concentration in EDA.

Together, these groups move the model beyond raw transaction fields toward features that describe *how unusual* a transaction and its surrounding metadata are — which is the same reasoning a manual fraud reviewer would apply.

08 Machine Learning Approach

Two classification models were built and compared: **Logistic Regression**, used as an interpretable baseline, and **Random Forest**, used to capture non-linear interactions between features.

Preprocessing Pipeline

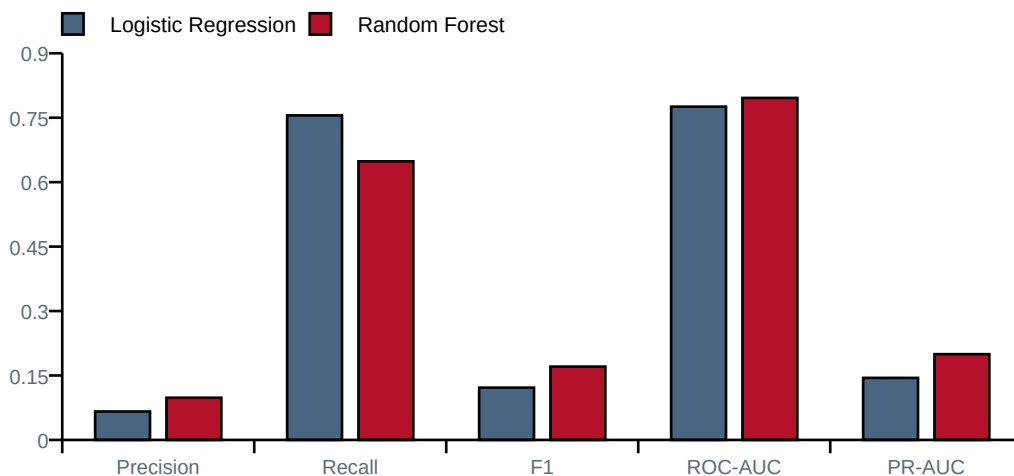
- Missing-value imputation applied separately to numerical and categorical fields.
- Numerical feature scaling for algorithms sensitive to feature magnitude.
- Categorical feature handling via one-hot encoding, with explicit handling of unseen/unknown categories at inference time.
- All preprocessing steps were wrapped in a single scikit-learn pipeline per model, so training and evaluation used identical, leakage-free transformations.

Both models were trained on the engineered feature set described in Section 7 and evaluated on a held-out split using the metrics in Section 9.

09 Model Evaluation

Because fraud is rare, accuracy is not reported as a meaningful metric. Instead, models are compared on precision, recall, F1, ROC-AUC, and **PR-AUC** — the last being the most informative metric for highly imbalanced classification, since it focuses on performance for the minority (fraud) class rather than being inflated by the large majority class.

Model	Precision	Recall	F1 Score	ROC-AUC	PR-AUC
Logistic Regression	0.0662	0.7554	0.1217	0.7757	0.1444
Random Forest	0.0982	0.6484	0.1706	0.7960	0.1996



Reading the Trade-off

- **Logistic Regression** achieved the higher recall (0.7554), catching more actual fraud cases, but at the cost of very low precision — flagging a large number of false positives.
- **Random Forest** achieved better precision (0.0982), F1 (0.1706), ROC-AUC (0.7960), and PR-AUC (0.1996), providing a more balanced trade-off between catching fraud and limiting false alarms.
- **Random Forest was selected as the best overall model based on PR-AUC**, since PR-AUC is the more reliable metric on this dataset's class imbalance and better reflects real-world investigation cost trade-offs than ROC-AUC alone.

This result should be read as a portfolio-scale comparison, not a claim of production readiness: the absolute precision and recall values indicate that further threshold tuning, feature refinement, or ensembling would be needed before any real deployment.

10 SQL & PostgreSQL Analysis

In addition to the machine-learning workflow, the transaction data was analysed independently in **PostgreSQL**, with the project connected directly to the database through the development environment. This SQL layer was designed to answer business-facing questions that a classification model alone does not directly surface.

Areas Covered

- **Data quality** — checks for missing values, duplicate records, and inconsistent field values across the transaction and identity tables.
- **Transaction analysis** — volume and amount patterns across products, cards, and time.
- **Behavioural analysis** — how transaction characteristics differ between fraud and non-fraud populations.
- **Risk segmentation** — grouping transactions by shared risk-relevant attributes (product, card, device/identity completeness) to support prioritisation.
- **Business questions** — translating the above into questions a risk or fraud-operations stakeholder would actually ask (e.g. which product/card combinations carry disproportionate fraud volume).

The key takeaway from this layer is that **machine learning identifies predictive patterns, while SQL helps translate transaction data into interpretable business and risk insights** — the two are complementary rather than redundant: SQL analysis is transparent and easily explained to non-technical stakeholders, while the ML model captures interactions that are harder to express as simple queries.

11 Risk Segmentation & Business Insights

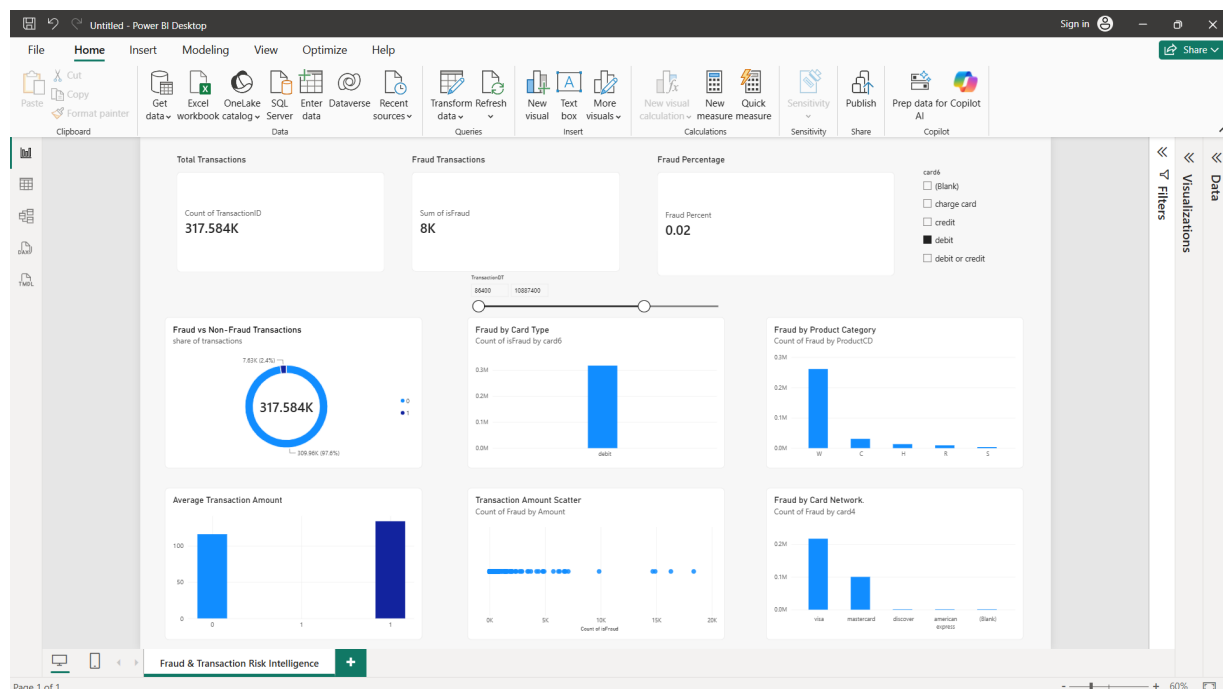
Bringing the modelling, SQL, and EDA work together, this section frames the technical results as insights a fraud or risk-operations team could act on.

- **Fraud is rare relative to legitimate transactions** (roughly 2.4% of volume), so detection is fundamentally a prioritisation problem, not a simple flag/no-flag decision.
- **Transaction amount and amount-deviation behaviour** provide a meaningful risk signal, reinforcing why amount-based features were engineered rather than using raw amount alone.
- **Missing identity or device information can itself be a risk signal** — incomplete metadata is not just a data-quality issue, but a pattern the model and SQL analysis both treat as informative.
- **Card and product characteristics reveal different fraud concentrations**, supporting segment-specific monitoring rather than a single uniform threshold across all transactions.
- **Risk segmentation helps investigators focus limited review capacity** on the transaction segments and individual cases most likely to be fraudulent, rather than treating all transactions equally.

This section is the analytical conclusion of the project: the combination of engineered features, model comparison, and SQL segmentation converges on the same message — fraud concentrates in identifiable segments, and a prioritisation-based approach is more realistic than a single “catch everything” classifier.

12 Power BI Risk Intelligence Dashboard

The Power BI dashboard consolidates the transaction and fraud data into an interactive view designed for business exploration rather than one-off analysis. It brings together volume, fraud rate, card, and product-level breakdowns on a single page, with date and card-type filtering for drill-down.



Power BI dashboard — Fraud & Transaction Risk Intelligence (screenshot from the live report; figures shown are the dashboard's own source-of-truth values).

What the Dashboard Communicates

- **Total Transactions, Fraud Transactions, and Fraud Percentage** as top-line KPIs for at-a-glance monitoring.
- **Fraud vs. Non-Fraud Transactions** donut view showing overall class balance.
- **Fraud by Card Type and Fraud by Card Network** to compare fraud concentration across card products (debit/credit, visa/mastercard/etc.).
- **Fraud by Product Category** to identify which product codes carry disproportionate fraud volume.
- **Average Transaction Amount and Transaction Amount Scatter** to compare spend behaviour between fraud and non-fraud transactions.
- **Date range and card-type filters** so a reviewer can interactively narrow the view to a specific period or card segment.

13 Project Architecture / Workflow

The project follows a linear, auditable pipeline from raw data to business-facing insight, implemented across six notebooks plus the SQL and Power BI layers.



Notebooks

#	Notebook	Purpose
1	Data Preparation	Load, join, and clean transaction and identity data.
2	Fraud EDA	Explore fraud distribution and behavioural patterns.
3	Feature Engineering	Build the 21-feature modelling set (Section 7).
4	Fraud Modelling	Train Logistic Regression and Random Forest pipelines.
5	Model Evaluation	Compare models using imbalance-aware metrics (Section 9).
6	Risk Analysis	Translate results into segmentation and business insights (Section 11).

14 Technology Stack

Technology	Purpose
Python	Data preparation, analysis, and modelling.
Pandas	Data manipulation and transformation.
Scikit-learn	Machine learning pipelines and preprocessing.
PostgreSQL	Database hosting and SQL-based analysis.
SQL	Data quality, behavioural, and risk-segmentation analysis.
Power BI	Interactive business intelligence dashboard.
Git / GitHub	Version control and project presentation.

15 Key Takeaways

- **Demonstrates a full analytics stack** — data preparation, feature engineering, machine learning, SQL, and BI — applied to a single coherent business problem rather than as isolated exercises.
- **Model results reflect the reality of imbalanced fraud data:** neither model is production-grade, but the comparison correctly shows Random Forest offering a stronger precision/recall balance than Logistic Regression.
- **Random Forest was selected on PR-AUC (0.1996)**, the metric most appropriate for rare-event detection — showing the ability to choose evaluation criteria that match the business problem, not just the easiest metric to optimise.
- **SQL analysis adds an interpretable, stakeholder-facing layer** that complements the ML model with transparent, queryable business logic.
- **Risk segmentation and the Power BI dashboard turn analysis into something actionable** — supporting investigation prioritisation rather than a black-box fraud/no-fraud label.

16 Conclusion

This project demonstrates a practical, end-to-end approach to fraud and transaction risk intelligence — moving from **data preparation** → **exploratory analysis** → **feature engineering** → **machine learning** → **SQL analysis** → **risk segmentation** → **business intelligence** in a single connected workflow.

It is presented as a portfolio project demonstrating applied analytical and machine-learning skills on a realistic, imbalanced fraud dataset — not as a claim of a production-ready fraud system. The combination of a compared, metric-appropriate ML model, independent SQL risk analysis, and an interactive Power BI dashboard reflects how a real analytics or risk team would structure this kind of investigation.